Why collective AI assurance cannot target agents or networks in isolation

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Abstract

AI governance increasingly targets coordination between agents rather than agents in isolation, but which feature of an interacting population produces a collective outcome has not been experimentally decomposed. We varied agent disposition, topology, interaction mechanic and peer sanctioning factorially across 264 runs of two multi-agent games, with matched non-LLM baselines and four model families. Structural and behavioural causes prove separable: a non-reasoning bot reproduces the public-goods game's degree–payoff relationship almost exactly, while in the trust game agents generate significantly more degree-linked inequality than a minimal reciprocal rule. Neither prosociality nor awareness prevented unequal outcomes here: agents following an identical fairness persona at rates from 84% to 100% across model families produced near-identical degree-linked inequality, and articulating one's own position conferred no payoff advantage. Most consequentially, whether disposition or topology explains the outcome inverts between the two interaction regimes, with model, personas and topologies held identical. Collective assurance therefore cannot generally be assigned to a fixed level of the system: the appropriate assurance target depends on the interaction regime and its causal structure.

1 Introduction

Populations of interacting large-language-model (LLM) agents can exhibit collective properties not predictable from any agent considered in isolation, and recent work has established this as a distinct class of risk requiring governance at the level of coordination rather than the individual system (Section 2). The concern is not hypothetical: health systems already deploy increasingly autonomous AI systems for administrative and coordination tasks such as preliminary triage and scheduling, while more autonomous medical agents remain largely at the development or pilot stage (Athni 2026). It is the coordination layer, not clinical autonomy, that is already in place — and that is the layer at which the properties studied here arise. Existing regulation is organised differently, assigning conformity obligations to identifiable systems and providers (European Union 2024) without supplying a substantive assurance criterion for distributional properties that arise between them. Alignment practice reinforces the individual framing, training and evaluating one model at a time (Ouyang et al. 2022; Bai et al. 2022).

Granting that the coordination layer is the right level at which to intervene leaves a question that literature does not settle: what *within* an interacting population actually produces the collective outcome, and does that locus stay fixed? The question has an immediate methodological edge, because an observed relationship between network position and outcome may be a structural property of the interaction rule rather than anything about the agents — Sheng et al. (2026) establish exactly such an effect in networks with no learned agents at all.

We therefore vary agent disposition, topology, interaction mechanic and sanctioning factorially, and run matched non-LLM baselines through the identical mechanic to separate what the structure guarantees from what the agents contribute. The central result is that no single governance target dominates: which factor explains the collective outcome depends on the interaction regime, inverting between our two environments under an identical model and identical personas.

Contributions. (1) A factorial separation of agent disposition, topology, interaction mechanic and sanctioning in two structurally distinct environments, with matched non-LLM baselines run through the identical mechanic — establishing that a high degree–payoff correlation is by itself no evidence of agent-specific behaviour, and locating where a real behavioural effect nonetheless appears. (2) Evidence that structural awareness in agents' stated reasoning is environment-dependent, and that where it is near-universal it still confers no payoff advantage once network position is controlled for. (3) A test of whether peer reputational sanctioning corrects the inequality it correctly targets, and a direct test showing that its measurable effect comes from the prompt cue rather than from the institution. (4) A variance decomposition showing that whether disposition or network position governs the collective outcome varies with the interaction regime — so that no fixed assurance target can be presumed adequate in advance.

2 Related work

2.1 Collective phenomena in LLM-agent populations

Building on work simulating societies of LLM agents (Park et al. 2023; Li et al. 2023), decentralised populations have been shown to develop shared conventions and acquire collective biases no individual agent exhibits (Ashery et al. 2025), while conformity dynamics can drive individually aligned agents into stably misaligned states (De Marzo et al. 2026). Piatti et al. (2024) find most LLM agents fail to reach sustainable equilibria in common-pool dilemmas; Han et al. (2025) that static network structure does not stabilise their cooperation as it does for humans; and de Arruda et al. (2026) that aggregate human-like outcomes can arise without matching individual mechanisms — a caution taken up in Section 4.2. Zhang and Krishnamurthy (2026) find preferential attachment and persistent centrality disparities when agents choose their own collaborators, where we assign topology exogenously so that position can be treated as morally arbitrary (Section 3.5). Outside stylised dilemmas, Cemri et al. (2025) locate much of the failure in deployed multi-agent frameworks in system design and inter-agent interaction rather than model capability. These studies establish that collective properties diverge from individual ones without decomposing why.

2.2 Network structure and distributional inequality

That individually reasonable local rules can produce unintended global patterns has long precedent: Schelling (1971) showed mild preferences over neighbours producing full residential segregation. Sheng et al. (2026) give the version that matters most here, showing theoretically and across empirical networks that allocation policies promoting cooperation concentrate resources on well-connected individuals, leaving peripheral ones worse off than under no cooperation at all; and the pattern appears in deployment, where Obermeyer et al. (2019) traced systematic bias in a healthcare allocation algorithm to the use of healthcare costs as a proxy for health need, illustrating how distributional bias can arise from system design without requiring discriminatory decisions at the point of allocation. Inequality of this kind is a candidate structural property of the allocation rule, not evidence about the agents — which is why our public-goods result is framed as a test against that baseline rather than a discovery, and why the non-LLM controls in Section 4.2 are load-bearing.

2.3 Governance beyond the individual agent

Hammond et al. (2025) organise multi-agent risks into a distinct class — miscoordination, conflict, collusion — underpinned by factors including network effects, and Tomašev et al. (2025) shift the unit of safety analysis from a single model to coordinated agent populations. Bohr (2026) makes the architecture concrete with *coordination transparency*: interaction logging, live monitoring, intervention hooks and boundary conditions, arguing that model-level assurance is necessary but insufficient. Pierucci et al. (2026) reframe alignment from preference engineering in agent-space to mechanism design in institution-space, and Bracale Syrnikov et al. (2026) show that a regime with enforceable consequences substantially reduces LLM collusion in Cournot markets while a prompt-only prohibition does not — a contrast our sanctioning result speaks to directly (Section 4.4). Earlier multi-agent work anticipated the design question (Shoham and Tennenholtz 1995; Boella and van der Torre 2006; Savarimuthu and Cranefield 2011); Shavit et al. (2023) translate it into engineering practice and Sarkar and Faik (2026) address the organisational logics shaping alignment decisions. This literature identifies the right *layer*; it does not supply an experimental decomposition of which feature within that layer drives the outcome.

2.4 Fairness and compositional assurance

Our use of “fairness” differs from the individual- and group-level criteria studied in the algorithmic-fairness literature (Dwork et al. 2012), which evaluate a single system’s output. We do not claim that our agents satisfy the formal individual-fairness criterion of Dwork et al. (2012), which constrains a classifier’s outputs relative to a task-specific similarity metric. Rather, every agent in the uniform-fair condition is given the same fairness-oriented disposition, and inequality is evaluated at the level of the resulting population — the finding being that uniformity of disposition is compatible with substantial collective inequality. Selbst et al. (2019) make the general point that fairness cannot be evaluated inside an isolated technical artefact abstracted from the system it sits in; this paper supplies a controlled instance of that abstraction failing.

The inference the paper tests — from component guarantees to a system property — is the object of assume-guarantee reasoning in compositional verification (Henzinger et al. 1998), where component guarantees hold only under explicit assumptions about the environment each component operates in — the framing that makes our result statable (Section 5).

3 Methods

3.1 Environments

Two structurally distinct multi-agent games, both built on a network-scoped local-payoff principle extending the local-neighbourhood mechanic of Nowak and May (1992). In the **networked public-goods game** (a commons dilemma in the tradition of Ostrom (1990)), each round an agent chooses to contribute a fixed amount, split equally among its visible network neighbours, or extract it for itself; payoff is network-scoped, so an agent’s degree is a real determinant of what it receives regardless of its own behaviour. In the **networked trust game** (medically framed: agents are treatment coordinators in a referral network), each round an agent chooses one visible colleague to send a trust-multiplied share of its resource to, or keeps it for its own patient — a targeted, dyadic mechanic whose outcome depends on real reciprocity rather than a symmetric split, per Berg et al. (1995). Both games were run rather than either alone because the public-goods game’s symmetric split makes an agent’s degree a partial mechanical determinant of its payoff by construction; the trust game’s targeted, dyadic structure has no such built-in guarantee, so a structural effect that appears in both environments cannot be attributed to one game’s specific payoff arithmetic alone. The two environments differ not only in the mechanic itself but in coupled properties of the interaction regime — payoff arithmetic, neighbourhood versus dyadic scope, and the role of reciprocity — a point taken up where regime differences are interpreted (Sections 4.6 and 5).

3.2 Population and manipulation

Alignment population (2 levels): uniform-fair (every agent under an identical fair/cooperative persona) or mixed (a fixed round-robin of fair, self-interested, and neutral personas). **Topology** (3 levels): fully-connected (null condition, no degree variance), clustered, and scale-free (hub-dominated). **Institution** (2 levels): a reputational peer-sanctioning mechanism off or on, inspired by the experimental punishment literature (Fehr and Gächter 2000; 2002) but differing from it in where the cost falls: a sanction lowers the target's reputation without altering any payoff, and is costly only to the sanctioner, whose call ends the turn and so forgoes that round's allocation (Supplementary Section S3). When on, the round prompt explicitly cues the sanctioning option from round 1. Population size $n = 20$, 15 rounds per run, 8 replicates per cell.

Replication was extended adaptively in a subset of cells, and the procedure is stated here in full so that it cannot be mistaken for undisclosed optional stopping. After the initial 8-run estimates had been computed, three public-goods population × topology conditions (uniform-fair × clustered, uniform-fair × scale-free, mixed × clustered) were extended to 20 replicates in both sanctioning arms — six cells, 72 additional runs. The criterion for extension was estimation precision, not direction of effect: in these cells the initial estimate of the H1 correlation or of the H3 Gini difference lay too close to zero or to the pre-specified equivalence margin to adjudicate the hypothesis, and a condition was always extended in both sanctioning arms symmetrically. The hypotheses, falsification criteria and analysis methods (Section 3.6) were fixed before the confirmatory dataset was analysed and were not modified by the extension decision; this prespecification was internal to the project, and the study was not preregistered. Analyses that require a balanced design use only the first 8 runs of every cell and are therefore unaffected by the extension: the variance decomposition (Section 4.6) and all H2 self-report results (Section 4.3). H1 and H3 estimates use all available runs, with run counts reported per cell (Table 1). The main study therefore comprises 264 runs. Contribute/extract (respectively send/keep) terminology is label-randomized per run. Behaviour is measured by the rate at which an agent takes the prosocial action each round: the *contribution rate* in the public-goods game and the *referral rate* in the trust game (sending rather than keeping). These rates, not agents' self-reports, are what the behavioural results in Section 4.1 report.

We deliberately do not call these rates "compliance". The fair persona for the trust game instructs the agent to refer *when it judges a colleague better placed* and not to hoard *by default* (Supplementary Section S1), so a round in which an agent keeps the resource after weighing its options is not necessarily a departure from the persona — agents that keep frequently cite network-level fairness as their reason for doing so. The referral rate is therefore a lower bound on persona-consistency, and differences in it between models may reflect how literally a model reads an instruction containing a judgment clause as much as how faithfully it follows one. The public-goods persona admits the same reading in principle, though there the contribution rate is at ceiling in every uniform-fair cell, leaving no room for the distinction to matter. Population size and round count were checked rather than assumed: a population-size sweep from 10 to 160 agents showed the core degree–payoff correlation and its precision essentially unchanged, and extending runs beyond 15 rounds did not alter the qualitative pattern, supporting $n = 20$, 15 rounds as an adequate, non-arbitrary operating point rather than a convenience choice.

3.3 Model and agent runtime

Each agent is defined entirely by prompt text, with no fine-tuning: a fixed persona statement for the whole run, plus a per-round message (a longer tutorial at round 1, a short state reminder thereafter). Each agent-turn consists of a read-only state check (visible neighbours, prior-round outcome, reputation signals) followed by exactly one action call, accompanied by a short free-text justification the agent supplies alongside its choice; this justification field is the raw material for the H2 self-report analysis (Section 3.4). When the sanctioning institution is inactive the action is contribute/extract or send/keep. When it is active, sanctioning is a third mutually exclusive option rather than an additional one: an agent that sanctions forgoes its resource decision for that round, which is recorded as no allocation and yields no payoff from its own endowment, though the agent still receives any incoming transfers. A turn ends programmatically as soon as the action call succeeds, rather than relying on a model-issued stop signal, since not every provider emits one reliably — necessary for consistent data collection across backends.

Primary model: DeepSeek (`deepseek-v4-flash`), a “lightweight” direct-completion runtime with full relevant context re-sent each round rather than a persistent stateful agent — the field-standard methodology for this kind of study (Han et al. 2025; Qiu 2025). This model carries the full factorial main study. Three further models from independent providers and architecture families (GLM-5.2, Mistral-Medium-3.5, Qwen3.5-122b) were run on the confirmatory cell only (uniform-fair $\times$ scale-free, sanctioning off, both environments) at 8 runs each — 48 additional runs — through the identical harness, prompts and game code (Section 4.5; per-backend configuration in Supplementary Section S7).

3.4 Metrics

Primary: Pearson correlation between an agent's network degree and its cumulative net payoff, computed *within* each replicate run and then averaged across runs, with 95% confidence intervals from bootstrap resampling over runs (10,000 resamples). The unit of independent observation is the run, not the agent: the 20 agents in a run share one generated network and one interaction history, so treating them as independent observations would understate uncertainty. Correlations pooled over all agent-runs in a cell are reported in Supplementary Information as descriptive statistics only. Gini coefficient of the final net-payoff distribution — a standard summary of inequality across a distribution, ranging from 0 (every agent ends with an identical payoff) to 1 (one agent holds the entire population's payoff); we report it as a second, distribution-level view of inequality alongside the degree–payoff correlation, which instead measures whether that inequality specifically tracks network position. Non-LLM baseline comparison (fixed, random, and, in the trust game, simple reciprocal strategies) run through the identical mechanic. H2: an LLM-judge classification, per agent-round, of self-reported reasoning against the agent's own visible feedback and its actual structural position, on two dimensions (situational awareness; fairness/trust assertion). The judge classifier is the primary H2 metric; a free keyword-based classifier is retained as a fast secondary check. Both were validated against blind human coding on a 30-item spot-check sample: agreement was substantial in the sense of Landis and Koch (1977) on both dimensions and against both automated classifiers ($\kappa = 0.66$–0.79). An initial human-coding pass on the awareness dimension was itself unusable (definition drift toward "contains any justification" rather than "references a specific structural relationship") and had to be redone with anchor examples added to the coding instructions — kept as a documented methods note, since it illustrates a real failure mode of unanchored human coding that the automated classifiers do not share.

3.5 What “unfairness” denotes here

The measured quantities — Gini and the degree–payoff correlation — establish *inequality* and its structural dependence, which are empirical facts. Calling that inequality *unfair* is a normative step and is licensed here only by a specific feature of the design: network position is assigned exogenously and at random, is not earned, and is uncorrelated with any agent's disposition, contribution, need, or performance — all of which are held identical by construction in the uniform-fair population. We therefore operationalize structural unfairness as *differences in final resource allocation attributable to exogenously assigned network position rather than to any difference in agent disposition, contribution, need, or performance.* Under a different normative criterion — one on which a hub legitimately earns more by bearing more coordination load — the same distributions would not count as unfair, and the paper's empirical claims would stand unchanged. We use "structural inequality" for the measurement and "structural unfairness" only where the above criterion is being invoked.

3.6 Hypotheses and falsification criteria

Three hypotheses were specified, with falsification criteria and analysis methods fixed before the confirmatory dataset was analysed (prespecified internally; the study was not preregistered — Section 3.2). H1 and H3 are stated with deliberately asymmetric scope: the confirmatory claim covers the uniform-fair $\times$ scale-free condition in both environments — the condition where structural advantage is largest and where the hypotheses are therefore most sharply testable — with all remaining cells reported descriptively.

H1 (structural inequality under uniform disposition). In a population of agents following an identical fairness-oriented persona, an agent's network degree predicts its cumulative net payoff. Disconfirmed if the 95% CI on the degree–payoff correlation in the confirmatory condition includes zero, or if the point estimate falls below $r = 0.30$.

H2 (self-report does not correct structural position). Agents' self-reported situational awareness does not prevent the structurally determined outcome. Evidence against H2 would be awareness language that is both prevalent and accompanied by an attenuated degree–payoff relationship.

H3 (peer reputational sanctioning does not reduce inequality). A reputational peer sanction — one that imposes a standing cost on the target but does not alter the resource ledger — does not reduce the Gini inequality of final payoffs, even where it demonstrably targets high-degree agents: its effect on Gini falls within a pre-specified practical-equivalence margin of ± 0.05. The hypothesis is deliberately restricted to this class of institution; it makes no claim about sanctions with enforceable economic consequences. Tested by bootstrap equivalence test (10,000 resamples, percentile CI) rather than parametric TOST. H3 is confirmed only if the CI lies entirely within the margin — a stricter standard than a non-significant difference.

3.7 Analysis model

The factorial structure is `outcome_metric` $\sim$ `alignment` $\times$ `topology` $\times$ `institution`, fit separately per environment, with run as the unit of observation. We report it as a fixed-effects variance decomposition on a balanced subset rather than as a fitted mixed-effects model; the reasons and consequences are stated where the decomposition appears (Section 4.6). Uncertainty on each variance share is a 95% percentile bootstrap interval over 10,000 resamples: replicates are drawn with replacement within each cell — the design's own resampling unit, alignment, topology and institution being fixed factors — and the identical closed-form decomposition is rerun on each draw, so interval and point estimate cannot diverge in their computation. All analyses use run-level resampling for inference and are implemented in pure-standard-library Python with no external numerical dependencies, so that every reported statistic is reproducible from the released raw exports without a specific solver stack.

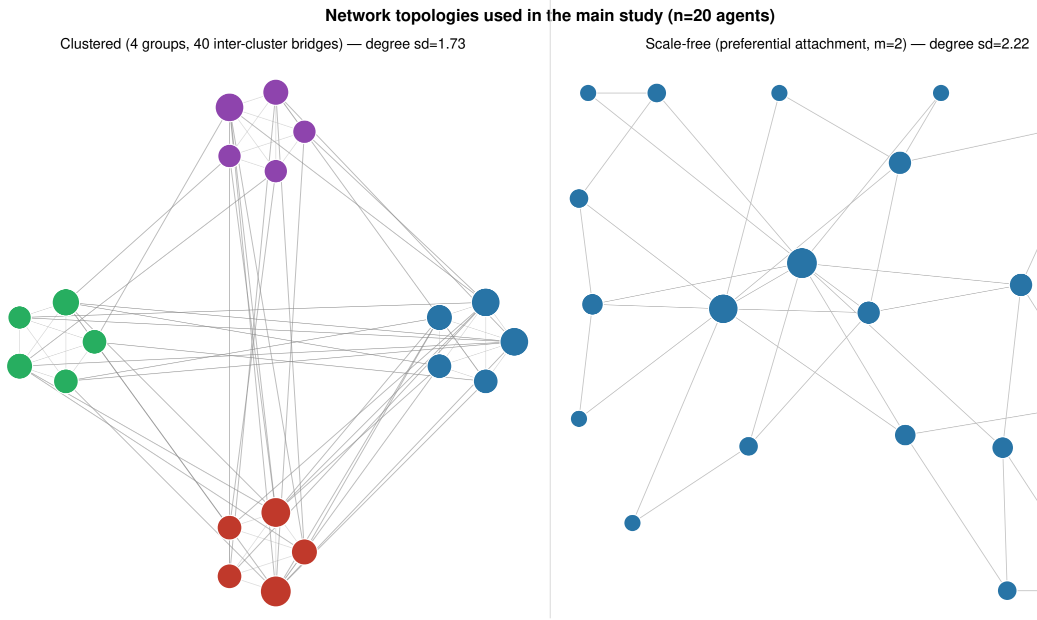

Figure 1: The two non-null network topologies used in the main study ($n = 20$ agents): clustered (4 groups, 40 inter-cluster bridges, degree s.d. = 1.73) and scale-free (preferential attachment, $m = 2$, degree s.d. = 2.22).

4 Results

4.1 H1: structural inequality under a uniform fairness disposition

Agents in the uniform-fair population took the prosocial action at markedly different rates across environments: a contribution rate of 100.0% in the public-goods game against referral rates of 80.0–89.6% in the trust game. The asymmetry matters for how the central claim reads: behaviour at ceiling occurs in the public-goods game, which is also where the structural effect is largely mechanically guaranteed (Section 4.2), while the trust game supplies the behavioural evidence at a lower rate. Neither environment alone carries the argument. Final payoffs nonetheless diverged sharply by network position in both. In scale-free topology, degree correlated strongly with cumulative net payoff in the public-goods game (mean within-run $r = 0.973$, 95% CI [0.968, 0.979] over 20 runs, sanctioning off) and comparably strongly in the trust game ($r = 0.967$ [0.958, 0.976], 8 runs). Gini coefficients were 0.538 [0.516, 0.560] and 0.526 [0.490, 0.561] respectively — substantial inequality generated purely by network position despite, in the public-goods game, essentially perfect individual behaviour. With `inter-cluster-edges` raised to 40 (Table 1, Fig. 1), clustered topology showed a comparably strong effect in the public-goods game ($r = 0.962$ [0.953, 0.970]), resolving the pilot-stage finding that the default bridge count gave clustered too little degree variance to correlate reliably; the trust game's clustered condition showed a real but substantially weaker effect ($r = 0.337$ [0.188, 0.493]) — degree still predicts payoff, but far less deterministically than in scale-free. Fully-connected topology, with no degree variance, showed no correlation by construction, though residual Gini there was non-trivial in the trust game (0.428–0.479) — the dyadic mechanic generates payoff variance without any degree advantage to exploit.

Round-level trajectories (Fig. 3) show this signature is near-immediate rather than accumulated: the public-goods correlation sits at its final value from round 1, and the trust game's rises from 0.839 to 0.972 by round 2 and remains essentially at that level for thirteen further rounds (0.967 at round 15) while raw inequality *falls* (Gini 0.818 → 0.526). Structural position is close to fully determinative from the first interactions, which distinguishes this from cumulative-advantage accounts where prominence compounds over time.

The mixed population showed a more environment-dependent pattern than anticipated. The structural effect survived heterogeneous disposition in both trust-game topologies ($r = 0.700$ and 0.306) and in scale-free public goods ($r = 0.340$), but collapsed in clustered public goods ($r = 0.095$ [−0.011, 0.200], 20 runs). Behavioural heterogeneity does not weaken the structural signal uniformly; it can suppress it almost entirely in some topology–environment combinations while leaving it intact in others.

Table 1: Mean within-run degree–payoff correlation (r) and Gini coefficient by condition, sanctioning off. CIs are bootstrap percentile intervals resampling over runs (10,000 resamples). Three public-goods population × topology conditions were extended to 20 runs (six cells in total, counting both sanctioning arms; Section 3.2); all others are at 8.

Environment	Population	Topology	runs	r [95% CI]	Gini [95% CI]
Public goods	uniform-fair	scale-free	20	0.973 [0.968, 0.979]	0.538 [0.516, 0.560]
Public goods	uniform-fair	clustered	20	0.962 [0.953, 0.970]	0.323 [0.298, 0.348]
Public goods	mixed	scale-free	8	0.340 [0.220, 0.456]	0.394 [0.380, 0.409]
Public goods	mixed	clustered	20	0.095 [−0.011, 0.200]	0.423 [0.411, 0.435]
Trust game	uniform-fair	scale-free	8	0.967 [0.958, 0.976]	0.526 [0.490, 0.561]
Trust game	uniform-fair	clustered	8	0.337 [0.188, 0.493]	0.367 [0.327, 0.407]
Trust game	mixed	scale-free	8	0.700 [0.617, 0.784]	0.375 [0.325, 0.426]
Trust game	mixed	clustered	8	0.306 [0.129, 0.483]	0.288 [0.272, 0.305]

4.2 Isolating the LLM-specific component

Non-LLM baselines were run through the identical mechanic and topologies at full scale (8 runs per cell) and compared against the real uniform-fair runs by two-sided permutation test (10,000 shuffles) on the difference in mean within-run r (all cells in Supplementary Section S8).

The comparison separates two claims. First, the correlation magnitude alone is close to a mechanically guaranteed property of the payoff structure once behaviour is uniform: a *fixed*-strategy bot, which does not reason at all, reproduces the real agents' public-goods correlation almost exactly (clustered: 0.960 vs. real 0.962, $p = 0.77$; scale-free: 0.982 vs. 0.973, $p = 0.09$). A high degree–payoff correlation is therefore not on its own evidence of anything LLM-specific — a negative control a study reporting only its own agents' correlations would miss.

Second, against that null, the trust game's reciprocal (tit-for-tat) comparison isolates a genuine behavioural effect: real agents produced substantially *more* degree-linked inequality than the minimal reciprocal heuristic, in both topologies and both sanctioning arms (scale-free, sanctioning off: $r = 0.967$ vs. 0.671, $p = 0.0001$; clustered: 0.337 vs. 0.018, $p = 0.026$). Agents that visibly track reciprocity do not converge on the equalizing behaviour a minimal reciprocal rule produces; their reciprocity concentrates around already-well-connected partners. The *fixed* baseline runs the opposite way here (scale-free: 0.982 vs. 0.967, $p = 0.014$; clustered: 0.708 vs. 0.337, $p = 0.0009$), mechanically cycling through neighbours yielding a *cleaner* relationship by removing the targeting variance real agents introduce. Real behaviour is reducible to neither heuristic.

4.3 H2: self-report versus structural position

Every agent-round free-text justification from the full 264-run dataset was classified by LLM judge (59,440 unique texts, deduplicated from 75,822 recorded justifications; zero classification failures). This includes the 72 runs added later to sharpen the H1 and H3 estimates, which were

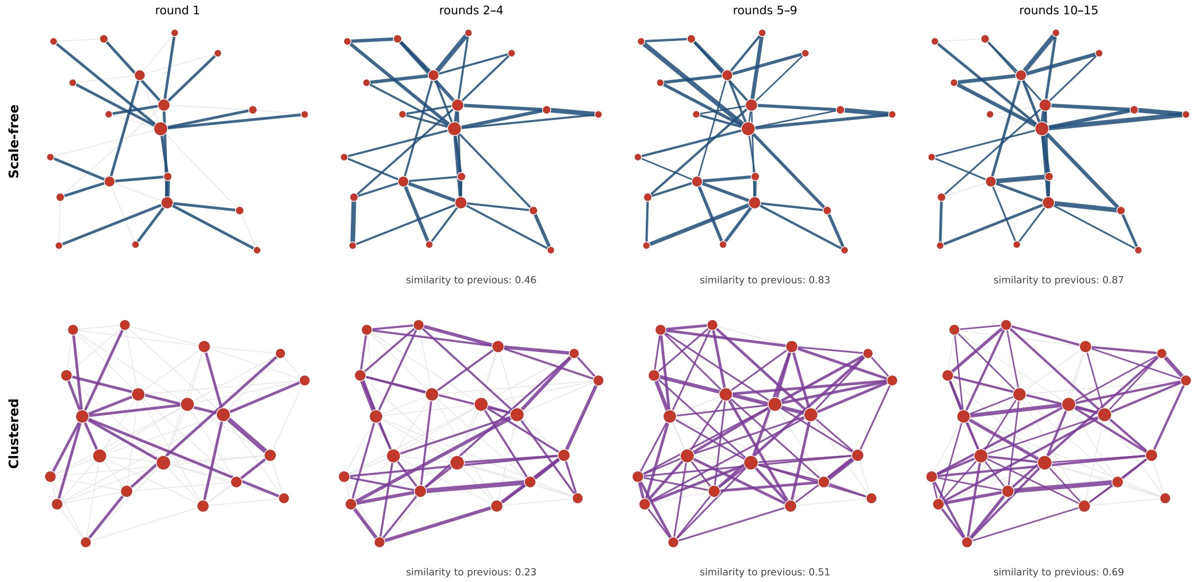

Figure 2: Referral dynamics in the trust game, uniform-fair population, sanctioning off; one representative run per topology. Each panel shows referrals occurring *within* the window indicated, not cumulatively: coloured edge width is proportional to referrals along that dyad in that window, grey lines are the underlying visibility graph, node size is degree. Values below each panel are cosine similarities between consecutive windows' referral vectors, averaged over all 8 runs in the cell. Scale-free referral structure reorganises once after round 1 and then holds ($0.46 \rightarrow 0.83 \rightarrow 0.87$); clustered structure keeps reorganising to the end ($0.23 \rightarrow 0.51 \rightarrow 0.69$), the condition in which the degree–payoff relationship is also weakest (Section 5).

reclassified under the same judge prompt as the balanced dataset, so the per-cell rates below rest on 8 runs per cell except in the three cells extended to 20.

Two things about agents' stated reasoning are worth separating. Invoking the collective is near-universal and mechanic-independent: essentially every agent in both environments frames its choice in terms of the network, fairness or trust (100.0% of public-goods justifications, 98.7% of trust-game ones, human-coded). Referring to one's own concrete position — naming a counterpart, a reciprocity history, one's own connectivity — is not. That is confined almost entirely to the trust game: a stratified, human-adjudicated sample (148 items, two blind coders, third-party adjudication of disagreements) gives a calibrated prevalence of 38.4% [18.8, 56.5] in the uniform-fair trust game against 0.0% [0.0, 0.0] in the public-goods game. An LLM judge applied to all 59,440 justification texts reproduces the same contrast on a broader criterion (68.9–81.4% against 0.1–0.2%; Supplementary Section S5). What an agent appears to "notice" about its structural situation is thus a property of the interaction regime, not of the agent's disposition — and the gap between environments is near-total on either measure.

This awareness did not translate into corrected outcomes. To test that directly rather than by juxtaposition, per-agent awareness rates were correlated with final payoff controlling for degree, computed within each run and averaged across runs (5,280 agent-replicate observations over all 264 runs, one per agent and run; CIs bootstrapped over runs). In the confirmatory condition — trust game, uniform-fair, scale-free — the partial correlation is -0.078 [-0.183, $+0.024$]: awareness confers no measurable payoff advantage once network position is accounted for.

Across the remaining cells the picture is mixed rather than uniformly null: small correlations of both signs appear ($+0.220$ clustered trust game, -0.171 mixed-population scale-free), none approaching the magnitude of the structural effect itself. The defensible claim is not that awareness is unrelated to outcome but that it is a weak and inconsistently-signed correlate of it while degree is a near-deterministic one. Awareness tracks disadvantage rather than advantage:

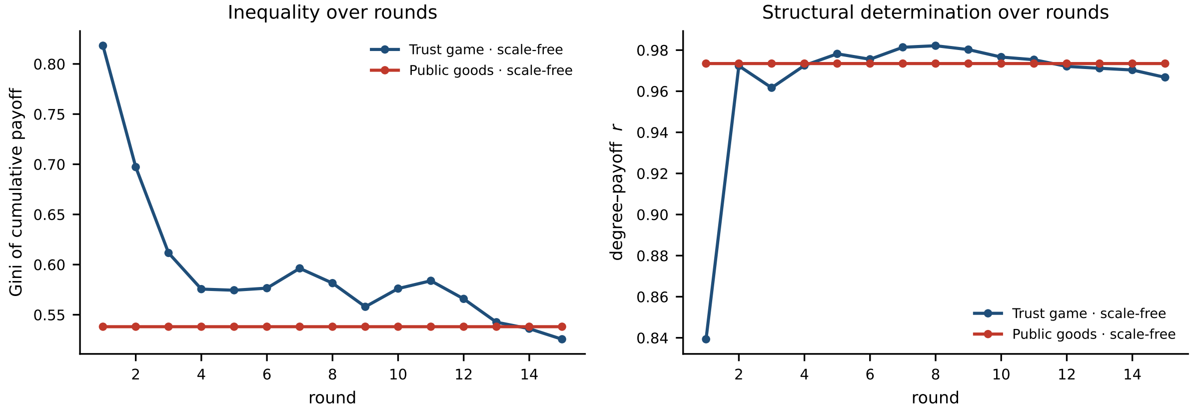

Figure 3: Round-level trajectories in the two central cells (uniform-fair × scale-free, sanctioning off), averaged over runs. *Left:* Gini of cumulative payoff. *Right:* degree–payoff correlation. The public-goods game sits at its final correlation from round 1, as its symmetric mechanic dictates; the trust game reaches its final level by round 2 and remains essentially there for thirteen further rounds, while raw inequality declines as early transaction sparsity resolves. Structural determination is thus near-immediate in both environments rather than cumulative — degree does not become more predictive of payoff as advantage compounds; it is close to fully predictive from the first interactions.

it correlates −0.184 with degree, so the structurally worse-off agents talk about their position most.

4.4 Institutional sanctioning

Sanctioning, once actively cued in the round prompt, was exercised at rates varying widely by cell (0.1–1.4 sanctions per run in fully-connected and clustered trust-game cells, up to 24.0–38.5 in mixed-population public-goods cells; Supplementary Section S9) and, where it was exercised more than negligibly, targeted high-degree agents in scale-free topology, where structural advantage is largest: in the three scale-free cells with a non-trivial number of sanctions, sanctioned agents averaged 1.59–2.08× the population mean degree, against roughly 1.0× elsewhere. The fourth scale-free cell (uniform-fair public goods) is the exception noted below, where a single sanction across 20 runs gives a ratio of 0.54× on which nothing rests. H3 was tested by bootstrap equivalence test against the pre-specified ±0.05 practical-equivalence margin in the two confirmatory cells (uniform-fair × scale-free, both environments), as set out in Section 3.6; the remaining cells are descriptive.

H3 is confirmed in one of its two confirmatory cells and resolved in the other by a follow-up experiment. In the trust game, ΔGini $= -0.003$, 95% CI $[-0.049, +0.043]$ — entirely inside the margin. In the public-goods game, ΔGini $= +0.046$, CI $[+0.012, +0.079]$ ($n = 20$ per arm): the interval excludes zero and its upper bound exceeds the margin, so the pre-specified test resolves that cell neither way.

The residual effect is not, however, produced by sanctioning. Agents in that cell issued *one* sanction across all 20 runs: public-goods agents contributing at ceiling leave no free-riding to punish, so the two arms differ only in that one prompt mentions an option never exercised. We tested this directly by adding a third condition in which the identical prompt clause appears but the sanctioning endpoint is disabled at the backend, so every attempt is rejected. All three arms are compared at $n = 8$ runs each — the cue-only arm's own replicate count, matched by the first 8 runs of the no-cue and full-sanctioning arms — so that arm means and differences rest on one basis. The cue alone reproduces the entire effect: cue-only Gini 0.606 [0.549, 0.662] against 0.532 [0.511, 0.553] with no cue ($\Delta = +0.073$, CI $[+0.015, +0.127]$), while adding real sanctioning power on top of the cue changes nothing detectable (0.576 [0.537, 0.614]; $\Delta = -0.030$, CI $[-0.091, +0.036]$). Announcing a symbolic institution measurably shifts behaviour here; enforcing it adds nothing detectable.

Across every cell, sanctioning never reduced inequality. Its main-effect share of variance in Gini is at or near zero throughout (Section 4.6), and no cell shows a negative ΔGini whose interval excludes zero. Where the institution was actually used — the trust game, where non-reciprocation gives agents something to punish — it correctly identified high-degree targets and left the distribution unchanged.

4.5 Cross-model robustness

The strong degree–payoff relationship is not specific to one backend. Three further models from independent providers and architecture families (Qwen3.5-122b, GLM-5.2, Mistral-Medium-3.5) were run through the identical harness on the central uniform-fair $\times$ scale-free, sanctioning-off cells in both environments, at the main study's replicate standard of 8 runs each. All four models land in $r = 0.95$–0.99 across both environments (Supplementary Section S11). These runs probe the central structural effect only: they do not test the full factorial design on the additional backends, and therefore do not establish that the disposition–topology inversion of Section 4.6 generalizes across model families (see Limitations).

The informative contrast is between individual behaviour and collective outcome. Trust-game referral rates vary by 16.2 percentage points across backends (DeepSeek 83.8%, GLM 96.6%, Qwen 99.9%, Mistral 100.0%) while the degree–payoff correlation varies by less than 0.03. Mistral referred in 2,399 of its 2,400 agent-rounds; DeepSeek kept the resource in 386 of 2,400, typically citing network-level fairness as its reason. The residual agent-rounds in each cell resolve to neither action — one for Mistral, two for DeepSeek — and are recorded as such rather than dropped (Supplementary Section S7). These are visibly different agent populations producing the same structurally determined outcome. Raising the rate at which agents individually do the prosocial thing — the lever an individual-system regime can actually pull — did not move the collective result in any of the four model families tested.

4.6 Variance decomposition

Variance was partitioned by closed-form fixed-effects ANOVA over cell and marginal means (Section 3.7), computed on the first 8 runs of every cell so that the design is fully balanced (Section 3.2); the rationale and the full table are given in Supplementary Section S10.1. This decomposition is the paper's core evidence for interaction-regime dependence.

It yields a result more consequential for governance than anticipated. There is no stable ranking of alignment against structure: *which factor dominates is itself a property of the interaction regime* (Supplementary Table S4). Every share below carries a percentile bootstrap interval obtained by resampling replicates within each cell and rerunning the same decomposition on each draw (10,000 resamples; Section 3.7). For the degree–payoff correlation, the public-goods game is population-dominated (population 85.5% [80.0, 91.2] against topology 2.6% [0.8, 5.4]) while the trust game inverts this almost exactly (topology 55.7% [45.0, 67.4] against population 9.8% [3.4, 20.0]). The inversion does not rest on point estimates alone: in both environments the two intervals are disjoint, and in opposite directions. For Gini, the trust game splits roughly evenly between population (35.9% [26.5, 46.7]) and topology (32.9% [22.9, 44.6]), while in the public-goods game a population $\times$ topology interaction is the largest single term (41.3% [23.0, 56.1]), though its interval overlaps that of the topology main effect (19.3% [8.9, 29.1]) and it therefore exceeds either main effect as a point estimate rather than separably.

Same model, same personas, same topologies: moving the population from the symmetric contribution environment to the targeted dyadic one reverses which factor explains the outcome. The two environments differ in coupled properties beyond the mechanic itself — payoff arithmetic, neighbourhood versus dyadic scope, the role of reciprocity (Section 3.1) — so what this establishes is dependence on the interaction regime as a whole, not an attribution to any single isolated property of it. The governance consequence is unchanged by that caveat, and it removes the natural fallback position for a single-system conformity regime: if structure reliably dominated, a regulator could certify structures instead, auditing topology to bound inequality; if disposition reliably dominated, per-model certification would suffice. The data show neither, so no fixed certification target can be presumed adequate in advance of knowing how the agents will be made to interact.

Two findings are robust across all four decompositions. Sanctioning's main-effect share is at or near zero throughout (0.0–0.4%), corroborating the H3 result; and in none does individual alignment account for outcomes on its own — either topology dominates outright, or the two interact too strongly to separate.

5 Discussion

Uniform prosocial behaviour at or near ceiling — the condition an individual-system conformity regime is built to certify — still produced large, structurally determined, unequal outcomes in both environments. The non-LLM baselines show part of this is close to graph-theoretically guaranteed once behaviour is uniform, which is itself part of the point: the failure is structural, not a property of any agent's disposition.

The awareness result is the sharpest form of this. Awareness of a structural problem, distributed across every individual in a population, is not the same thing as any individual being able to act on it: in the condition where most justifications referenced the agent's own position, awareness carried no payoff advantage once position was controlled for (Section 4.3).

Read together, the three correctives that fail here map onto the components a coordination-layer governance architecture is built from. Bohr (2026) proposes monitoring interaction and providing intervention hooks; the results locate where each stops short. Awareness is observability without control: agents in the trust game saw and articulated their position, and the population-level outcome was unaffected (Section 4.3). Sanctioning is intervention without causal leverage: the institution identified the structurally advantaged agents correctly and left the distribution that made them advantaged untouched (Section 4.4). Uniform disposition is the component-level guarantee itself, held at ceiling to no effect (Section 4.1). Observing a coordination problem and being able to act somewhere in the system are both necessary; neither is sufficient without leverage over the mechanism that generates the outcome.

This is where our result and Bracale Syrnikov et al. (2026) agree from opposite directions. In their Cournot markets a regime with enforceable consequences substantially reduces collusion while a prompt-only prohibition does not bind at all. Our sanction sits on the prompt-only side: it is reputational, and the cue-only experiment shows the announcement carries whatever effect there is. The conclusion is not that institutional governance fails, but that an intervention may correctly identify the agents responsible and still change nothing if it does not reach the mechanism generating their advantage — here, the resource ledger rather than the reputation ledger.

The governance implication needs care. This study does not show that these agents would pass a legal conformity assessment; following a fairness persona is not legal conformity. The claim concerns the *unit* of assessment rather than its stringency. An assessment confined to each agent independently — however demanding — could not detect the effect shown here: every agent-level property it might examine is held constant across conditions that differ substantially in collective outcome, so it has no variable to bind on.

Stated formally, the implicit step is one of *compositional assurance*: that establishing agent-level conformity C for every component suffices to establish a collective property F of the system they form,

$$\bigwedge_{i=1}^{n} C(a_i) \;\Rightarrow\; F(I(a_1, \ldots, a_n))\,.$$

C and F sit at different levels: C is an agent-level behavioural guarantee — conformity to a fairness norm, evaluable on one agent — approximated here by the observed prosocial-action rate, which in the trust game is a lower bound on persona-consistency rather than a direct measure of it (Section 3.2); F is a distributional property no agent can satisfy or violate alone. Write the interaction regime as $I = (A, T, M, G)$: agent dispositions, topology, mechanic, governance.

Our results are a counterexample to the unconditional implication. Holding every $C(a_i)$ at or near ceiling leaves F free to vary with I alone. This is not news to compositional verification, where component guarantees have long been understood to hold only relative to explicit assumptions about each component's environment (Henzinger et al. 1998). The productive form of the claim is therefore the assume-guarantee one,

$$\left[\bigwedge_{i=1}^{n} C(a_i)\right] \wedge A(I) \;\Rightarrow\; F(I(a_1, \ldots, a_n))\,,$$

with $A(I)$ the conditions on the interaction regime under which agent-level conformity does compose. The open question is not whether composition works, but *which assumptions on I it requires* — and our results bear directly on how hard that question is.

They show, first, that F is not a function of A or T separately in either of our environments but of the regime as a whole, $F = f(A, T, M, G)$; and second, more restrictively, that even the relative weight of A and T changed with M (Section 4.6). Any candidate $A(I)$ quantifying only over agents, or only over topology, therefore cannot be presumed adequate in advance of knowing the mechanic — which also closes off the obvious repair for an individual-system blind spot, certifying the structure instead. Recent frameworks correctly identify the coordination layer as the governance target (Bohr 2026; Pierucci et al. 2026); these results add that the layer is not homogeneous, and its internal causal structure must be identified before an intervention can be aimed at it.

An unresolved anomaly. One condition resists this account: in clustered topology the trust game shows only a weak degree–payoff relationship ($r = 0.337$, against 0.967 in scale-free), and its referral structure never stabilises (Fig. 2). We report it as an open question; the supporting measurements and a testable conjecture are set out in Supplementary Section S10.2.

The claim these results support can be stated compactly. Agent-level conformity does not compose unconditionally into collective fairness: holding every agent's prosocial behaviour at or near ceiling leaves the collective outcome free to vary with the interaction regime alone. That much is a counterexample to an inference assurance regimes make implicitly. What makes it hard to repair is the second half: *collective assurance cannot generally be assigned to a fixed level of the system in advance; the appropriate target depends on the interaction regime and its causal structure.*

Limitations. These are simulated agent populations, and the network-scoped payoff formula is one reasonable operationalisation of structural access advantage rather than the only one. The two environments differ in several coupled properties — payoff arithmetic, neighbourhood versus dyadic scope, the role of reciprocity — so the design identifies dependence on the interaction regime; it does not isolate which property of the regime carries the inversion, and it establishes a counterexample to unconditional compositional assurance rather than an impossibility theorem extending beyond the regimes tested. The cross-model evidence covers four architectures but only the two central cells, so whether the environment-dependence in Section 4.6 also holds across backends is untested. The H2 measure characterises what agents *say* about their position: free-text justifications are not guaranteed to reflect the computation that produced the action, only a plausible narration of it (Turpin et al. 2023), and prevalence estimates depend on a classification scheme whose validation we report in Supplementary Section S5.

Data availability

Code, configuration, and exported round-level data are publicly available at `https://github.com/Entropy-Hackers/collective-AI-assurance`. Full persona and round-prompt text, tool schemas, non-LLM baseline strategy definitions, the H2 classification pipeline, topology generation, and model/API configuration for every backend tested are provided in the Supplementary Information.

Use of large language models

Large language models were used as a tool in this work: for implementing the simulation harness and analysis code, for computing reported statistics, and for drafting and revising manuscript text. All experimental design decisions, all interpretations of results, and all claims made in this paper were determined and verified by the authors, who reviewed every generated output and take full responsibility for the content. Large language models are not authors of this work. Their use as the experimental subject of the study, as distinct from their use as a tool, is described in Section 3.

Competing interests

The authors declare no competing interests.

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Author contributions

H.M. conceived the study, coordinated it, and drafted the manuscript. H.M. and A.H. developed the conceptual framework. M.P. implemented the simulation and performed the analysis. A.H. reviewed and edited the manuscript. All authors approved the final version.

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